

Reforming EU Electricity Market Design via PPAs, CfDs, and Long-Term Signals: Evidence and Policy Options from Italy

Francesco Decarolis^{*,†} and Tommaso Pigliacampo[†]

April 2026

Abstract

This paper discusses how the EU electricity market design reform can strengthen long-term investment signals while limiting the downsides of extreme price volatility. It focuses on the coexistence between two families of long-term instruments: decentralized power purchase agreements (PPAs), which can reveal value and mobilize demand, and public contracts for difference (CfDs), which can provide revenue stability but may crowd out private contracting if poorly designed. The paper develops a risk-allocation perspective to explain why markets remain incomplete (e.g., exposure to price, shape, volume, basis, credit, and regulatory risk) and maps these frictions to a set of policy options. It proposes an implementation-oriented toolbox combining: (i) information and standardization measures to reduce transaction costs and improve valuation; (ii) targeted public de-risking through guarantees and credit support; (iii) aggregation and public procurement to mobilize credible long-term demand; and (iv) auction and scheme design choices that preserve incentives for bilateral PPAs alongside CfDs. Italy is used as an illustrative case to show how these instruments interact in practice and to derive EU-relevant design principles and sequencing recommendations.

Keywords: Electricity market design; Long-term contracting; Variable Renewable Energy; Contracts for Difference (CfDs); Power Purchase Agreements (PPAs); Risk allocation; Energy policy

JEL Codes: G13, L51, L94, Q41, Q48

^{*}Department of Economics, Bocconi University; ENEL Foundation Chair in Competitiveness and Transition

[†]BAFFI Centre, Bocconi University